

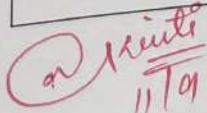
PROJECT AND TEAM INFORMATION

Project Title

TravelMate Uttarakhand – Tourism Recommendation System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)**Motivation (1 pt)**

Travel Mate Uttarakhand – Tourism Recommendation System is a C based application designed to check users to discover stable tourist destination according to their budget, stay time, season and interest in Uttarakhand.

The project uses various data structure algorithms, including array, stack, queue, linked list, sorting, graphs and searching. A recommendation score is calculated for different destination, and the most suitable and affordable destination are showcased/displayed to the user. The graph structure can also be used to represent connection between tourist destination and to find suitable routes.

The main objective of this project is to combine data structure with practical real world tourism application making travel planning easier, while demonstrating the practical concept of data structure using C.

State of the Art / Current solution (1 pt)

Currently, Uttarakhand tourism information is provided through mobile apps and website that offer detail about destinations, activities, routes, and travel services.

The current tourism system mainly focuses on providing information and travel services. However, a student-oriented system can improve the experience by providing personalised recommendations based on budget, interests, season, number of days, and travel preferences. The Uttarakhand tourism policy itself proposes making destinations discovery easier by allowing users to search destinations based on interest and activities.

Limitations of current solutions

- *Information is spread across different sections/platforms.*
- *Users may need to manually compare multiple destinations.*
- *Personalized recommendations based on budget + season + interest are limited.*
- *Finding the best destination for a specific student/ traveler profile can take time.*
- *Route and destination selection can be improved using data structure and algorithms.*

Our project, "Travel Mate Uttarakhand – smart tourism recommendation system" addresses these limitations by using graph, searching, sorting, linked list, stack and queues to generate personalized tourism recommendation.

Project Goals and Milestones (2 pts)

Project Goals

- Develop a Uttarakhand tourism recommendation system using C.
- Recommend destinations based on budget, interest, and season.
- Use data structure like arrays, linked lists, stacks, queue and graphs.
- Provide searching, sorting, and route-friendly features.
- Create a simple and user-friendly interface.

Project Milestones

- Provides personalized tourist recommendations in Uttarakhand.
- Helps users select destinations based on budget, season, interest and duration.
- Demonstrate practical use of Data Structure in C.
- Makes travel easier and faster.
- Improves understanding pf real-world application of DS and algorithms.

Project Approach (3 pts)

- **Collect Data** – Gather information about Uttarakhand tourist destinations, budget, seasons and activities.
- **Store Data** – Use structure and array to store destination details.
- **Apply Data Structure** – Use linked lists, stacks, queues, and graphs for managing tourism data and routes.
- **Recommendation** – Match user performance and calculate a recommendation score.
- **Search & Sort** – Search destinations and rank them according to their scores.
- **Display Result** – Show the top recommendation destinations and suitable routes to the user.
- **Testing** – Test the system with different user preferences and improve accuracy.

System Architecture (High Level Diagram)(2 pts)

PROJECT PROPOSAL



Project Outcome / Deliverables (1 pts)

- Provides personalized tourist recommendation in Uttarakhand
- Helps users select destinations based on budget, season, interest, and duration.
- Demonstrates practical use of Data Structure in c.
- Provides search, sorting, and route-finding features.
- Makes travel planning easier and faster.
- Improves understanding of real-world applications of DS and algorithms.

Assumptions

- Tourism destination data is predefined and stored in the system.
- User provides correct information about budget, season, interest, and duration.
- Destination details such as cost, category, and rating are assumed to be accurate.
- Internet connectivity is not required for the basic system.
- Recommendations are generated using predefined rules and Data Structures.
- The system is designed mainly for educational/project purposes.

References

1. Uttarakhand tourism – Official Website – Destination and tourism information.
2. GeeksforGeeks – Data Structure and Algorithms in C.
3. W3Schools – C Programming and Data Structures.
4. Programiz – C Programming and Data Structures.
5. Tourist Care Uttarakhand – Tourism and travel information.